Unsupervised Dual-Path Consensus Machine Learning and Operational Policy Framework for Village Fund Expenditure Anomaly Detection

AUTHORS

Farrell Yodihartomo • Humam Faiq • Pandu Wicaksono • Charlie Yen

Bina Nusantara University & Corruption Eradication Commission (KPK), Jakarta, Indonesia

01 / PROBLEM BACKGROUND

Rural Oversight Deficit

Fiscal Scale

Rp 71 T

Distributed across 75,259 villages annually.

State Losses

Rp 598.13 B

591 court convictions recorded Rp 598.13 billion lost.

Reactive Oversight

2–5 Years

Cases adjudicated 2–5 years after budget disbursement.

Auditor Shortage

5–15 Auditors

Only 5–15 auditors oversee 250 regency villages.

Swakelola Dominance

98.8%

98.8% of Jambi activities bypass competitive bidding.

02 / LITERATURE GAPS (2022–2026)

Three Core Research Gaps

GAP 01

Subspace Cancellation

Majority voting erases local density isolates.

GAP 02

Missing Policy Mapping

Outlier scores lack statutory audit rules.

GAP 03

Geographic Cost Distortion

Mountain logistics inflate baseline unit costs.

03 / DESIGN SCIENCE RESEARCH (DSR)

Anomaly Detection Pipeline

STAGE 01

Siskeudes Hygiene

96,778 clean records across 1,363 villages.

STAGE 02

Regional Centering

27 engineered features using regency z-scores.

STAGE 03

Tri-Engine Inference

Parallel execution across three orthogonal engines.

STAGE 04

Audit Checklist Mapping

Feature error decomposition into field checklists.

04 / ALGORITHM & FEATURE DESIGN (RQ1)

Tri-Engine Mathematical Formulation

STAGE 01

Regional Baseline Z-Score Centering

$$z_{i,c,k,t} = \frac{x_{i,c,k,t} - \mu_{c,k,t}}{\sigma_{c,k,t} + \varepsilon}$$

ENGINE 01

Isolation Forest (IF)

200 trees isolate global sparsity at 90th quantile.

ENGINE 02

Local Outlier Factor (LOF)

20-neighbor local density ratio at 95th quantile.

ENGINE 03

Reconstruction Autoencoder (RDA)

8-layer neural compression (3.4:1) at 95th quantile.

05 / ENSEMBLE INNOVATION (RQ2)

Dual-Path Consensus Gate

Protocol 1: Majority Voting (≥ 2 Engines)

Subspace Orthogonality: IF-LOF Cohen's Kappa reaches only 0.041.

Mutual Cancellation: Eliminates 3,940 high-risk local density isolates.

Detection Failure: Misses 49.0% of injected synthetic fraud cases.

Protocol 2: Dual-Path Consensus

Gating Formula: LOF OR (IF INTERSECT RDA).

Local Channel (LOF): Preserves heavy-tailed density isolates (BC = 0.957).

Global Channel (IF ∩ RDA): Suppresses single-model noise via dual convergence.

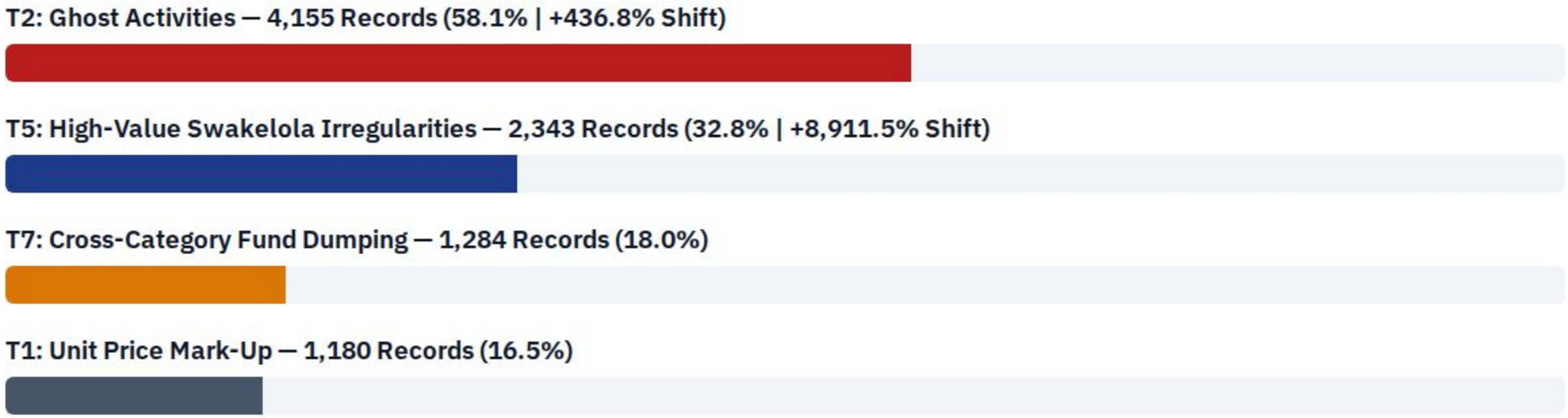
06 / EXPERIMENTAL EVALUATION (RQ2)

Detection Model Ablation Study

Model Configuration	Precision	Recall	F1 Score	AUC-ROC	Jambi Flags	Realization Risk
LOF Alone (Density)	0.686	0.686	0.686	0.745	4,839 (5.0%)	Rp 396.40 Billion
IF Alone (Sparsity)	0.724	0.724	0.724	0.782	9,678 (10.0%)	Rp 451.18 Billion
RDA Alone (Neural Loss)	0.812	0.812	0.812	0.811	4,840 (5.0%)	Rp 428.60 Billion
Global Intersection (IF ∩ RDA)	0.854	0.640	0.732	0.815	2,314 (2.4%)	Rp 246.45 Billion
Protocol 1 (Majority ≥ 2)	0.720	0.510	**0.568**	0.642	3,107 (3.1%)	Rp 278.40 Billion
Protocol 2 (Dual-Path)	**0.846**	**0.846**	**0.846**	**0.912**	**7,153 (7.4%)**	**Rp 642.85 Billion**

07 / OPERATIONAL POLICY MAPPING (RQ3)

Corruption Typology Shift



08 / XAI ERROR DRIVERS & PERSISTENCE

Error Drivers & Village Tiering

Primary Indicator

42.7%

Category Cost Deviation

Primary error driver across 2,065 records.

Secondary Indicator

32.0%

Activity Unit Cost

Secondary error driver across 1,551 records.

Persistence Flag

702

3-Year Persistent Villages

51.5% of villages flagged every year.

Regional Density

15.8%

Batanghari Regency

Highest risk density at Rp 115.71 billion.

09 / INFORMATION SYSTEMS SUCCESS IMPACT

Inspectorate Audit Efficiency

92.6%

Audit search space reduction for district inspectorates (APIP)

Filters 96,778 Siskeudes ledger entries down to 7,153 high-risk targets paired with automated field inspection checklists.

10 / LEGAL GOVERNANCE & LIMITATIONS

Evidentiary Law & Boundaries

Initial Investigative Lead

Algorithmic flags guide preliminary inquiry.

Administrative Irregularities

High scores require mens rea verification.

Audit Chain of Custody

Feature errors direct physical site audits.

Off-Book Cash Limits

Unrecorded cash kickbacks remain undetectable.

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THANK YOU • Q&A SESSION

Conclusion & Key Findings

MODEL PERFORMANCE

0.846 F1

AUC = 0.912

Dual-Path Consensus

Achieves high predictive reliability and robustness in detecting financial anomalies.

RISK ISOLATION

Rp 642.85B

Realization Exposure

Financial Risk Mitigation

Isolates substantial realization risk across budget lines for targeted intervention.

AUDIT EFFICIENCY

92.6%

Search Space Reduction

APIP Inspectorate Impact

Drastically reduces manual audit workload, directing resources to high-risk targets.

